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MATH 245A

Discussion 2
Due:

1. Review

Problem 1.1

Let $(X, \mathcal{A}, m)$ be a finite measure space.
(a) If $A_1 \subseteq A_2 \subseteq \cdots$ are measurable, then $m\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{i\to\infty} m(A_i)$.
(b) If $A_1 \supseteq A_2 \supseteq \cdots$ are measurable, then $m\left(\bigcap_{i=1}^{\infty} A_i\right) = \lim_{i\to\infty} m(A_i)$.

Solution.
1. Suppose $A_1 \subseteq A_2 \subseteq ...$ are measurable. Note that

$$\mu[A_2] = \mu[A_1 \cap A_2] + \mu[A_2 - A_1].$$

Assume the inductive hypothesis,

$$\mu[A_n] = \mu[A_1] + \sum_{k=1}^{n} \mu[A_{k+1} - A_k].$$

By the use definition of measurable, and we see

$$\mu[A_{n+1}] = \mu[A_1] + \sum_{k=1}^{n+1} \mu[A_{k+1} - A_k].$$

Thus

$$\begin{aligned}
\mu\left(\bigcup_{k=1}^{\infty} A_k\right) &= \mu\left(A_1 \cup \left(\bigcup_{k=1}^{\infty} (A_{k+1} - A_k)\right)\right) \\
&= \mu[A_1] + \sum_{k=1}^{\infty} \mu[A_{k+1} - A_k] \quad \text{disjointness} \\
&= \lim_{n\to\infty} \left(\mu[A_1] + \sum_{k=1}^{n} \mu[A_{k+1} - A_k]\right) \\
&= \lim_{n\to\infty} \mu[A_n]
\end{aligned}$$

2. $A_1 \supseteq A_2 \supset ...$ measurable: Take complements and use a.

2. The rules of measurable sets

Problem 2.1

Scholium. If $P(n)$ is a statement about the nth natural number, how do you (hope to) prove $P(n)$ is true?

Solution.

Problem 2.2

Let $\mathcal{B}$ be the Borel σ-algebra on $\mathbb{R}$. Suppose m, n are two measures on $(\mathbb{R}, \mathcal{B})$ with $m((a,b)) = n((a,b)) < \infty$ for every $-\infty < a < b < \infty$. Prove $m(A) = n(A)$ for all $A \in \mathcal{B}$.

> *Solution.* This is an appliction of $\pi - \lambda$ theorem. Let $P = \{(a,b) : -\infty < a < b < \infty\}$. P is a π system. Let $L = \{B : B \text{ Borel}, \mu(B) = \eta(B)\}$. $P \subseteq L$. We will show that L is a λ-system.
> 1. $\mathbb{R} \in L$: Let $A_k = (-k, k)$. $A_1 \subseteq A_2 \subseteq ... \subseteq A_k \subseteq ....$
>
> $$\mathbb{R} = \bigcup_{n=1}^{\infty} A_n \Rightarrow \mu(\mathbb{R}) = \lim_{n\to\infty} \mu(A_n) = \lim_{n\to\infty} \eta(A_n) = \eta(\mathbb{R})$$
>
> 2. $A, B \in L$ where $B \subseteq A$ then $A - B \in L$:
>
> $$\eta(B) + \eta(A - B) = \eta(A) = \mu(A) = \mu(B) + \mu(A - B) \Rightarrow \eta(A - B) = \mu(A - B)$$
>
> 3. Growing unions are the same logic as 1.
>
> Thus L is a λ-system. Furthermore, by $\pi - \lambda$ theorem, $\mathcal{B} = \sigma(P) \subseteq L$. Hence we are done.

Problem 2.3

let $(x, \mathcal{A})$ be a measurable space, $\boxed{?} \subseteq \mathcal{A}$, and m, n two σ-finite measures on $(x, \mathcal{A})$ with $m(a) = n(a)$ for all $a \in \boxed{?}$.
(a) is it true that $m(a) = n(a)$ for all $a \in \sigma(\boxed{?})$, the smallest σ-algebra containing $\boxed{?}$?
(b) what if m, n are finite measures?

> *Solution.*

Problem 2.4

Work in $X = [0, 1]$ with the usual topology and Lebesgue measure m.
(a) Prove that for every Borel $A \subseteq X$ and every $\varepsilon > 0$ there are a compact K and a relatively open U with $K \subseteq A \subseteq U \subseteq X$ and $m(U \setminus K) < \varepsilon$. Prove this from scratch using only basic properties of m.
(b) Deduce that every Borel $A \subseteq X$ has an F_σ subset F and a G_δ superset G with $F \subseteq A \subseteq G$ and $m(G \setminus F) = 0$.

> *Solution.*
> 1.

3. Quantifiers and measurability

Problem 3.1

Let $f_n : \mathbb{R} \to \mathbb{R}$ be a sequence of functions. Describe $\left\{x : \lim_{n\to\infty} f_{n(x)} \text{ exists}\right\}$.

> *Solution.*

Problem 3.2

Let $f : \mathbb{R} \to \mathbb{R}$ be arbitrary. Prove $C_f = \{x \in \mathbb{R} : f \text{ is continuous at } x\}$ is a G_δ set, hence Borel. *Hint:* for each $k \geq 1$, take the union of all open intervals I with $|f(y) - f(z)| < 1/k$ for all $y, z \in I$.

> *Solution.*

Problem 3.3

Let $A \subseteq \mathbb{R}$ be Lebesgue measurable, and define

$$E_\infty = \{x : x + 1/n \in A \text{ for infinitely many } n \geq 1\}, \quad E_{\text{ev}} = \{x : x + 1/n \in A \text{ for all sufficiently large } n\}.$$

(a) Show both are Lebesgue measurable. If A is Borel, are they Borel?
(b) Prove $E_{\text{ev}} \subseteq E_\infty$, and give an example where the inclusion is strict.

> *Solution.*

Problem 3.4

If $E_1, E_2, \cdots \subseteq [0, 1]$ are Lebesgue measurable and $\sum_{i=1}^{\infty} m(E_i) < \infty$, what can you say about $m(\{x : x \in E_i \text{ infinitely often}\})$?

> *Solution.*

Problem 3.5

(Ternary digits.) For $x \in [0, 1)$ write $x = \sum_{n=1}^{\infty} d_n(x)/3^n$ with $d_{n(x)} \in \{0, 1, 2\}$, using the expansion ending in zeros when there are two choices.
(a) Fix a word $(e_1, \cdots, e_N) \in \{0, 1, 2\}^N$. Show that the set of numbers whose first N digits are $e_1, \cdots, e_N$ is a half-open interval, and find its endpoints.
(b) Prove that $B = \{x \in [0, 1) : \lim_{N\to\infty} \#\{1 \leq n \leq N : d_n(x) = 2\}/N = 1/2\}$ is Borel.
(c) Show the set of $x \in [0, 1]$ admitting two different expansions has Lebesgue measure 0.
(d) What is $K = \{x : d_n(x) = 0 \text{ or } 2 \text{ for all } n \in \mathbb{N}\}$? What is its Lebesgue measure?

> *Solution.*

Problem 3.6

Suppose A is Lebesgue measurable and $B = \bigcup_{x\in A}[x - 1, x + 1]$. Prove B is Lebesgue measurable.

> *Solution.*

4. Two interesting problems

Problem P1

Construct a Borel set $A \subseteq \mathbb{R}$ such that for every interval (a, b),

$$0 < m(A \cap (a, b)) < |b - a|.$$

> *Solution.*

Problem P2

(Qual problem.) Let $\tau > 1$ and define

$$E_\tau := \{x \in \mathbb{R}^2 : \exists C_x > 0 \text{ such that for all } \lambda \in \mathbb{Z}^2, |x \cdot \lambda| \geq C_x / \ |\lambda|^\tau\}.$$

Prove E_τ has full Lebesgue measure.

> *Solution.*